Shaz (Ethan Shahrooz) Moghaddam

Data Scientist · Python Developer

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Right to work: British citizen · DBS check available on request

PROFILE

Data scientist and Python developer with a Data Science Bootcamp qualification from Imperial College London and HyperionDev (96% average). I take data from raw files to tested, deployed products: cleaning and analysing it, modelling it with scikit-learn and NLP, and shipping it with FastAPI, SQL and Docker. I also publish open-source tools, including a Python charting library on PyPI. I'm looking for a data science or ML engineering role where I can own work from the first question to production.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

Data and ML: pandas, NumPy, SciPy, scikit-learn, TensorFlow (LSTM forecasting), spaCy, NetworkX, Pydantic

Methods: linear and logistic regression, decision trees, random forests, K-means clustering, anomaly detection (Isolation Forest), causal discovery (PC algorithm), NLP, time-series analysis

LLMs and AI tools: Claude API for structured extraction and plain-English questions about data; Claude, ChatGPT and Gemini used daily in development

Data and APIs: PostgreSQL, SQLite, SQLAlchemy 2.0, Alembic, FastAPI, Flask, REST APIs, ONS API, OAuth 2.0 (Auth0)

Visualisation: Matplotlib, Seaborn, Plotly Dash, Streamlit, Tableau, React, bezel-charts (my own library)

Engineering: Docker, Git, GitHub Actions (CI and scheduled workflows), pytest, Playwright, Jupyter, Render, Python packaging and PyPI trusted publishing

SELECTED PROJECTS

Jobs & the Economy, Open UK Labour Market Challenge

Python · pandas · bezel-charts · ONS API · HMRC PAYE payroll data · GitHub Actions · pytest · MIT licence

- Public data challenge on UK jobs, unemployment and pay. One script downloads 33 ONS labour market series and HMRC payroll data, and a report of eight findings is generated from the data, so every number and sentence stays true as new figures arrive.
- Analysis covers long-term sickness (up 620,000 since Covid to 2.74 million), youth unemployment (15.6%), vacancies per unemployed person, real pay since 2001, a Phillips curve (r = −0.65, 2001 to 2025) and where survey and payroll data diverge.
- A weekly GitHub Actions job tests the data and opens a pull request with refreshed charts after each ONS release. Builds are identical across pandas versions, enforced in CI. All 15 charts are drawn with my own library.

github.com/ShazMoghaddam/uk-jobs-economy

Bezel Charts, Open-Source Chart Library for JavaScript and Python

JavaScript · Python · Web Components · SVG · PyPI · pytest · Node test runner · Playwright · MIT licence

- Ten zero-dependency chart types as web components (about 12 kB gzipped), and as a Python package on PyPI that draws them from pandas DataFrames for notebooks, reports and READMEs.
- The Python renderers are a line-by-line port, tested byte for byte against the JavaScript output across 119 chart cases and 689 number-formatting cases, enforced in CI. Taken to 1.0 in five releases, published from Git tags via PyPI trusted publishing.
- A no-code Studio imports CSV, Excel or JSON and exports HTML, a shareable link or a static SVG. Accessible to screen readers.

github.com/ShazMoghaddam/bezel-charts · pypi.org/project/bezel-charts · Live demo

House Prices and the Economy, Open UK Data Challenge

Python · pandas · Matplotlib · HM Land Registry UK House Price Index · GitHub · MIT licence

- Public monthly data challenge on GitHub analysing how UK house prices respond to economic shocks, built on the HM Land Registry UK House Price Index, with six reproducible charts and a pandas starter script.
- Time-series analysis of peak-to-trough falls and recovery times by region after 2008 (London about 4 years, the North East 13.5, Northern Ireland still below its 2007 peak), the stamp duty holiday spike, and the 4% dip as Bank Rate rose from 0.1% to 5.25%.

github.com/ShazMoghaddam/uk-house-prices-economy

LedgerLet, Making Tax Digital Platform for UK Landlords

Python · FastAPI · PostgreSQL · SQLAlchemy · Alembic · React · TypeScript · Auth0 · Docker · GitHub Actions

- Full-stack platform that keeps landlords' property records and files cumulative quarterly updates to HMRC through its Making Tax Digital APIs. Runs end to end on HMRC's sandbox.
- Bank statement CSV import with a two-step preview, column mapping and duplicate detection by row fingerprints. Money is stored as integer pence, with joint-ownership shares applied by a tested rounding rule.
- Connects to HMRC over OAuth with encrypted tokens and fraud prevention headers, keeps an append-only audit trail, and is checked by 394 backend tests on SQLite and PostgreSQL plus a CI gate with Playwright browser and accessibility tests.

ledgerlet.onrender.com (live demo) · Source code available on request

VoltEdge, Energy Intelligence Platform

FastAPI · Plotly Dash · scikit-learn · NetworkX · SQLite · Docker · Claude API

- Monitors energy use across four industrial sites using 90 days of demo data, with anomaly detection for each site (Isolation Forest) and root-cause analysis through causal discovery (PC algorithm), scored by causal strength.
- Evaluated the detector on a 30-day time-based hold-out with labelled anomalies. Found that outlier cleaning was hiding the largest spikes; fixing it raised recall from 76% to 94%, cut false alarms by 58% and doubled F1.
- ESG reporting (Scope 2, GRI 302-1, CSRD), a Claude assistant for plain-English questions, JWT login with role-based access and 911 passing tests. Deployed with Docker on Render.

github.com/ShazMoghaddam/voltedge · Live demo

VoltScope, Energy Bill Extraction and Validation

FastAPI · Claude API · Pydantic · SQLite · Docker

- Reads UK commercial energy bills from PDFs and phone photos (including iPhone HEIC) and extracts them into a fixed Pydantic schema using the Claude API. Also works as the bill module of VoltEdge.
- Rule-based validation checks MPANs, reconciles subtotals and VAT, and flags estimated readings, duplicate charges and contract renewal dates. Exports JSON, Excel and CSV, checked by 50 passing tests based on real anonymised bills.

github.com/ShazMoghaddam/voltscope

ClinIQ, Clinical Trial Site Performance Dashboard

FastAPI · Plotly Dash · SQLAlchemy · scikit-learn · spaCy · Claude API · Docker

- Built for mid-sized CROs to compare trial sites on enrolment speed, risk and protocol deviations: analyses deviation reports with spaCy, scores site risk with scikit-learn, and includes a Claude assistant for plain-English questions about trial data.

github.com/ShazMoghaddam/cliniq

SalaryAxis, UK Salary Benchmarking Platform

Python · Flask · React · pandas · NumPy · SciPy · Docker · ONS open data

- Salary benchmarking on ONS ASHE data with CPI inflation adjustment, regional and occupation breakdowns, gender pay gap trends and CAGR. Python REST API and React dashboard, with multi-stage Docker builds for AWS ECS Fargate or GCP Cloud Run.

github.com/ShazMoghaddam/salaryaxis

EXPERIENCE

Independent Product and Software Developer 2022 – Present

Remote · Google Play Store · Gumroad, Lemon Squeezy and Envato

- Built and published seven Android and Flutter apps on Google Play, owning each one from first idea through to release.
- Sell digital products including finance tools, generative art and screensavers, improving them based on customer feedback.
- Published Orbs, an open-source JavaScript library of ten morphing dot animations for loaders and AI "thinking" states: one file of about 5 KB with no dependencies, supporting reduced motion and keyboard use (github.com/ShazMoghaddam/orbs).
- Built the data and AI projects listed above during this period.

Career Break Mar 2013 – 2022

Full-time primary carer for my two children.

- Project-managed a loft conversion and kitchen extension without a main contractor: council approvals, sourcing and scheduling trades, and the budget. Finished in about 10 months for around £140k, against quotes of £180k to £200k.

Sales Representative (Technology) Mar 2011 – Mar 2013

John Lewis and Currys (via agency) · London

- Advised customers on technology products across several London stores, building communication and prioritisation skills.

EDUCATION AND CERTIFICATIONS

Data Science Bootcamp, Imperial College London and HyperionDev June 2025

96% average. Covered SQL and databases, data cleaning and preprocessing, statistics, visualisation (Tableau and Python), supervised learning, K-means clustering and NLP, finishing with three capstone projects.

The Data Science Course: Complete Data Science Bootcamp (Udemy, 32 hours) Oct 2025

iOS and Swift: The Complete iOS App Development Bootcamp (Udemy, 60.5 hours) Jan 2024

100 Days of Code and The Complete Python Bootcamp (Udemy, 58 hours) Oct 2022 – Jul 2023

Google Play Store Listing Certificate (Google Play Academy) Apr 2026

ADDITIONAL

Writing: Technical articles on dev.to (dev.to/shazmoghaddam) and seven personal essays on Medium (medium.com/@shaz.moghaddam).

Open source: Open-source work selected for the XERJ Corpus Hub (hub.xerj.org/contributors/ShazMoghaddam).

Availability: Full-time, freelance or contract. On-site, hybrid or remote in London, and open to international opportunities.